PART 1: CURRICULUM OVERVIEW & PEDAGOGICAL ALIGNMENT

An NSF RET Computational Materials Science Integration Framework

🏛️ Section 1: Executive Summary & Vision

Traditional introductory physics curricula often present the physical world as an idealized collection of perfectly frictionless planes, rigid massless strings, and invariant springs. While these abstractions are vital for teaching foundational vector dynamics, they inadvertently create a conceptual chasm between secondary school physics and authentic, cutting-edge engineering research. In the real world, structural components are not idealized geometric constraints; they are dynamic, multi-body quantum-mechanical systems that deform, fatigue, and fracture.

This unit bridges that gap by transforming abstract principles from **AP Physics C (Mechanics)** and **AP Calculus (AB/BC)** into an immersive, hands-on computational materials science investigation. Over a multi-week trajectory, students step into the role of software engineers and materials scientists to build a **Molecular Dynamics (MD) Simulation Engine** completely from scratch within a Google Colab environment using Python.

By scaling their models from a primitive two-atom interaction up to a 3D crystalline matrix and ultimately to a heterogeneous, multi-element atomic chain, students discover how macroscopic engineering observables—such as **Stress, Strain, Hooke's Law, Young's Modulus, and Ultimate Tensile Strength**—are emergent phenomena arising directly from the calculus of microscopic interatomic potential fields. The unit culminates in an open-ended engineering challenge: decoding the physics of fracture mechanics to understand why structural materials preferentially rupture at phase boundaries and interfaces.

💡 Section 2: Pedagogical Rationale & The Power of Simulation

1. The Limitations of Physical Labs in Materials Science

In a traditional high school lab setting, testing the mechanical properties of materials requires destructive testing equipment (such as hydraulic tensile testers or Charpy impact hammers). These apparatuses are not only cost-prohibitive for most schools, but they also function as a "black box." Students pull a metal specimen until it breaks, look at a digital readout of the force, and generate a macro-scale plot.

The underlying atomic mechanisms remain entirely invisible. Students cannot see the electron cloud overlaps, they cannot track the instantaneous bond lengths, and they cannot witness the individual potential well destabilizations that trigger the macro-scale tear.

2. The Computational Paradigm Shift

Computational modeling and simulation redefine the boundaries of the classroom lab. By constructing their own MD simulation engine, students gain omniscient visibility into the atomic landscape. They are empowered to:

- **Isolate Variables with Mathematical Perfection:** Strip away structural macroscopic impurities, air resistance, and complex thermal noise to observe pure, deterministic physical laws in action.
- **Access Non-Observable Physical Quantities:** Explicitly track and plot properties that are physically unmeasurable in a real-world high school lab, such as the exact instantaneous force vector between any two interior atoms.
- **Conduct Rapid Virtual Prototyping:** Instantly change the fundamental chemical identity of their material system by altering a few lines of code (e.g., swapping out the potential parameters of Aluminum for Copper or Titanium) and observe the systemic macro-stiffness response in milliseconds.

This approach honors the core objective of the **National Science Foundation (NSF) Research Experiences for Teachers (RET) program**: inspiring the next generation of innovators by replacing passive recipe-based learning with authentic computational experimentation.

🎯 Section 3: Detailed Lesson Architecture & Storyline

The curriculum is structured as a tightly scaffolded three-tier progression, designed to systematically build student confidence in both mathematical physics and programming syntax:

Tier 1: The Microscopic Pairwise Foundation (2-Atom System)

Students begin their journey by focusing on the simplest atomic building block: two isolated, neutral metal atoms. Students unpack the dual nature of matter—discovering how electrostatics create a fierce short-range repulsion balanced by a long-range Van der Waals attraction.

- **The Calculus Core:** Students apply the power rule and the chain rule to derive the analytical interatomic force function as the negative spatial derivative of the Lennard-Jones potential ($F = -dU/dr$).
- **The Coding Milestone:** Students translate their hand-derived calculus formula into functional Python syntax. They generate a vector array of separation distances to map out the non-linear force landscape and mathematically identify the roots of the function to pinpoint the stable equilibrium spacing coordinate (r_0).

Tier 2: The Meso-Scale Continuum Bridge (32-Atom 3D Lattice Matrix)

Next, students scale up their engine to handle a multi-body crystalline system. Utilizing professional molecular dynamics techniques, they assemble a 32-atom Face-Centered Cubic (FCC) crystal block of Aluminum.

- **The Linear Algebra Transition:** Students shift from managing simple variables to operating on multi-dimensional data grids (**matrices**). They utilize distance matrix algorithms to resolve hundreds of pairwise atomic interactions simultaneously.
- **The Materials Science Emergence:** Students code a virtual tensile testing loop that applies directional strain (ε) to the entire crystal coordinate matrix. By extracting the collective internal restoring forces across the pulling boundaries, they plot an engineering stress-strain curve. They apply a linear regression model to the elastic window to extract the slope, discovering how microscopic bond curvatures scale up to dictate a bulk material's macroscopic stiffness parameter: **Young's Modulus (E)**.

Tier 3: The Macro-Scale Engineering Grand Challenge (Heterogeneous Interface Fracture)

The curriculum culminates in an advanced fracture mechanics simulation. Students design a 1-dimensional wire characterized by an abrupt material phase boundary interface—joining a stiff, high-strength atomic domain with a soft, compliant domain.

- **The Critical Misconception Trap:** Students confront a deep conceptual trap regarding Newton's Laws. They must grapple with the fact that under static tension, the force transmitted through every single link in the chain must be perfectly uniform ($\sum F = 0$). This invalidates the common misconception that materials break at boundaries due to a "force concentration."
- **The Rupture Analysis:** Students apply a constant external tensile load to the wire within a dynamic time-integration environment. They observe the runaway acceleration of specific coordinates as bonds are pushed past their mathematical inflection point, discovering that fracture location is a race condition governed entirely by localized **strain tolerances, pre-strains, and ultimate strength thresholds ($F_{\max}$)**.

📋 Section 4: Comprehensive NGSS Standards Mapping

This curriculum is explicitly designed to satisfy the rigorous dimensions of the **Next Generation Science Standards (NGSS)** for high school physical sciences:

1. Performance Expectations (PEs)

- **HS-PS1-3 (Matter and its Interactions):** *Plan and conduct an investigation to gather evidence to compare the structure of substances at the bulk scale to infer the strength of electrical forces between particles.*
  - **Unit Connection:** Students utilize their simulation engine to alter interatomic potential parameters (ϵ and σ). They directly investigate how changing these microscopic electrostatic energy depths systematically scales up to modify bulk macroscopic physical constants (Young's Modulus) and material behaviors (fracture).
- **HS-PS2-6 (Motion and Stability: Forces and Interactions):** *Communicate scientific and technical information about why the molecular-level structure is important in the functioning of designed materials.*

- **Unit Connection:** The Tier 3 Grand Challenge forces students to analyze and defend how localized spatial variations in atomic spacing and material properties create structural weak points. Students write technical lab reports communicating how engineering failure modes are dictated entirely by molecular-level parameters.

2. Science & Engineering Practices (SEPs)

- **Practice 2: Developing and Using Models:** Students do not interact with the simulation as a static animation or a video game. They active author, modify, and test the code, moving from a simplified 1D pairwise model to a complex multi-body 3D matrix. They learn how mathematical constraints and cutoffs are applied to mirror real-world physical systems.
- **Practice 5: Using Mathematics and Computational Thinking:** This is the heart of the curriculum. Students use calculus derivatives to establish force fields, utilize vectorized matrix math to solve multi-body spatial distances, implement numerical integration techniques (Velocity Verlet) to track kinematics over time, and apply linear statistical regressions to determine engineering material properties.

3. Crosscutting Concepts (CCCs)

- **Scale, Proportion, and Quantity:** Students actively work across different dimensions. They observe lengths measured in Angstroms (10^{-10} m) and energies in electron-volts (10^{-19} J), and they code mathematical scale conversions to output macroscopic engineering data measured in Gigapascals (10^{9} Pa or Newtons per square meter).
- **Stability and Change:** Students map out the explicit transition from stable elastic equilibrium ($\sum F = 0$, where displaced atoms vibrate but retain structural integrity) to dynamic, unstable, irreversible change (fracture mechanics, where an inflection threshold is crossed and the material ruptures).

NGSS Standards Alignment

- **HS-PS1-3 (Matter and its Interactions):** Plan and conduct an investigation to gather evidence to compare the structure of substances at the bulk scale to infer the strength of electrical forces between particles.
- **HS-PS2-6 (Motion and Stability):** Communicate scientific and technical information about why the molecular-level structure is important in the functioning of designed materials.
- **Science & Engineering Practices (SEPs):** Practice 2 (Developing and Using Models); Practice 5 (Using Mathematics and Computational Thinking).